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Beyond Commonalities: A Quantitative Perspective on Syntactic Features across Mandarin Chinese Varieties

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ABSTRACT

While Mandarin Chinese varieties show notable syntactic differences, existing research has relied primarily on experiential reflection or corpus-based approaches, with a lack of data-driven investigations. This study presents a syntactically annotated corpus that includes Chinese Mainland Mandarin (CMM), Singapore Mandarin Chinese (SM), and Indonesian Mandarin Chinese (IM). Quantitative analyses of sentence length, dependency distance, and their distributions across the three Mandarin Chinese varieties demonstrate that: (1) average sentence length and mean dependency distance differ significantly across the three varieties. CMM shows the longest sentences and the highest mean dependency distance, SM occupies an intermediate position due to sustained Chinese education policies, and IM, shaped by historical discontinuities in Chinese education, ranks lowest on both measures; (2) the sentence length distributions of all three varieties conform to the Extended Positive Negative Binomial distribution, while their dependency distance distributions align with the right-truncated modified Zipf-Alekseev distribution, indicating shared cognitive mechanisms in language processing; (3) the distributional parameters of both quantitative linguistic models effectively distinguish the three Mandarin Chinese varieties, providing further support for the applicability of quantitative linguistic laws and their cross-linguistic typological universality. This study provides new approaches and perspectives for both Mandarin Chinese varieties and quantitative linguistics.

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1. Introduction

With the growth of overseas Chinese communities, distinct geographical varieties of Mandarin Chinese have gradually emerged. The linguistic differences among these Mandarin Chinese varieties, similar to those in World Englishes (Collins, 2009), have been widely recognized and extensively

researched (e.g. Lin et al., 2019; Tang, 2018; Yip & Tang, 2022). Overseas Mandarin Chinese varieties have not only been shaped by contact with English, local languages, and Chinese dialects, but have also evolved differently due to political, economic, cultural, and social divergences across regions (Huang et al., 2013). For example, in Southeast Asia, the use of Mandarin Chinese reflects markedly divergent language policies across different regions. In some countries such as Singapore (Chin & Cavallaro, 2021), it has been actively promoted, whereas in others such as Indonesia (Setijadi, 2015), it experienced prolonged periods of suppression. As a result, the trajectories of Mandarin Chinese heritage vary considerably, making these regions valuable sites for observing the sociolinguistic variation of Mandarin Chinese across overseas Chinese communities.

In recent years, scholarly interest in overseas Mandarin Chinese varieties has been steadily increasing. However, most existing studies have concentrated on localized analyses of specific grammatical variations, relying primarily on experiential reflection or corpus-based approaches (Jiang, 2018). While this line of research has provided valuable insights, it remains constrained by subjectivity, limited comparability, and insufficient replicability. These limitations are particularly salient in the study of grammatical variation, whose inherent complexity makes it difficult to identify generalizable patterns (Jiang, 2018). In contrast, quantitative research can capture and measure syntactic complexity at the macro level, thereby providing evidence that is more objective, replicable, and generalizable. This perspective is consistent with Bresnan and Hay (2008), who argue that grammatical variation is often characterized by subtle and gradient differences that are best revealed through quantitative approaches. The adoption of such methods helps to overcome the shortcomings of traditional introspective approaches, which often fail to capture large-scale systematic differences, and therefore provides a stronger methodological foundation for the study of Mandarin Chinese varieties. Nevertheless, quantitative investigations of Mandarin Chinese variation remain scarce, and existing corpora offer only limited coverage. Furthermore, most comparative studies to date have focused on contrasts between overseas Mandarin varieties and Chinese Mainland Mandarin (CMM), with relatively little attention to horizontal comparisons among overseas Mandarin Chinese varieties themselves. These limitations have significantly constrained our comprehensive understanding of Mandarin variation across diverse sociolinguistic contexts.

As mentioned above, this study focuses on two Southeast Asian Mandarin Chinese varieties, specifically Singapore Mandarin Chinese (SM) and Indonesian Mandarin Chinese (IM), with CMM serving as a reference. A data-driven approach is adopted to investigate similarities and differences in their macro-level syntactic complexity. Based on a syntactically annotated corpus of over one million words, we employ Average Sentence Length (ASL) and Mean

Dependency Distance (MDD) as the two key measures of syntactic complexity. This study also examines whether the distributions of sentence length and dependency distance conform to established quantitative probabilistic models, and evaluates the discriminative power of model parameters in distinguishing among different Mandarin Chinese varieties.¹ The findings are expected to offer empirical evidence for Mandarin Chinese variation research and to promote the broader application of quantitative linguistic methods in this domain.

2. Literature Review

2.1. *Inspirations from World Englishes Research*

With the advance of globalization, English, as a global lingua franca, has gradually developed into distinct localized varieties with specific features (Melchers et al., 2019). Since the 1980s, research focusing on World Englishes and related issues has steadily emerged, producing significant findings about the similarities and differences among English varieties across linguistic levels (Collins, 2009; Kranich et al., 2020; Lange & Leuckert, 2019; Revis & Bernaisch, 2020; Rohdenburg & Schlüter, 2009). The concentric circle model proposed by Kachru (1985) offers an effective framework for representing the global distribution and functional stratification of English varieties. Over the past five decades, the integration of corpora into World Englishes research has proven pivotal, with corpus analysis emerging as an important methodological approach (Collins, 2008; Grieve, 2011). Quantitative methods have also evolved from basic descriptive analyses, such as frequency comparisons (Schmied & Hudson-Ettle, 1996; Schneider, 2004), to more sophisticated techniques, including multidimensional and cluster analysis (Biber, 1987; Xiao, 2009), multifactorial logistic regression modelling (Heller et al., 2017), and confirmatory approaches used to predict the linguistic choices of speakers of different varieties of World Englishes (Deshors & Bernaisch, 2019; Gries & Bernaisch, 2016). Building on these developments, Variation-Based Distance and Similarity Modelling (VADIS) (Szmrecsanyi & Grafmiller, 2023) further extends the variationist paradigm by systematically measuring grammatical distance between varieties of World Englishes based on probabilistic grammar modelling. In addition, there are numerous well-established and readily accessible large-scale corpora available for the study of World Englishes, such as the International Corpus of English (ICE; Greenbaum, 1991) and the Global Web-based English Corpus (GloWbE; Davies & Fuchs, 2015).

2.2. *Research on Mandarin Chinese Varieties and Its Limitations*

Lexical differences have been the earliest and most extensively studied aspect of research on Mandarin Chinese varieties, whereas grammatical studies

have emerged relatively recently. Zhu (2016) authored the first monograph on grammatical variation in the Singapore Mandarin variety, while the six-volume Global Chinese Grammar stands as the first comprehensive multi-volume work on Mandarin Chinese varieties (Kuang, 2022). To systematically understand the diffusion and variation of Mandarin Chinese in overseas contexts, scholars have drawn on the ‘Three Concentric Circles’ model from World Englishes research and proposed the framework of the ‘Three Concentric Circles of Global Chinese’ (Goh & Lim, 2009). Goh and Lim (2009) categorize Mandarin Chinese used in the Chinese Mainland and the Taiwan region as belonging to the ‘Inner Circle’, Mandarin Chinese learned as a heritage language in overseas Chinese communities as the ‘Outer Circle’, and Mandarin Chinese learned as a foreign language in non-heritage contexts as the ‘Expanding Circle’. This framework offers a preliminary map of the global Chinese linguistic ecology. At the same time, linguists have increasingly called for moving beyond administrative boundaries and political affiliations as primary criteria for classifying language varieties (Xu & Liu, 2020). Instead, they advocate for a language-centred approach to constructing a more explanatory and inclusive taxonomy of Global Mandarin, thus promoting a paradigm shift from policy-oriented to usage-based perspectives. Although previous studies have established a solid foundation for research on Mandarin Chinese varieties, significant gaps remain in the field.

First, although the distinctive linguistic features of overseas Chinese publications are widely recognized, research on overseas Mandarin Chinese varieties has largely depended on intuitive judgements and observational analysis (Jiang, 2018). With the development of Mandarin Chinese variety corpora, quantitative methods have been introduced, but studies mainly focus on micro-level grammatical phenomena such as verb-object compounds (Jiang & Huang, 2024), light verbs (Huang et al., 2014; Xu et al., 2022), aspect marking (Khoo & Lin, 2018), *yǒu*+VP construction (Khoo & Lin, 2019; Teo, 2022), and theme-recipient alternation (Li et al., 2024), while lacking macro-level investigations of grammatical variation. This gap stems from the absence of a mature linguistic analysis framework for large-scale overseas Mandarin corpora and a lack of quantitative methods for overall linguistic features. Quantitative linguistics provides a new perspective for studying Mandarin Chinese varieties objectively at a macro level, with recent studies confirming its effectiveness (Hou & Huang, 2020; Hou, Huang, Ahrens, et al., 2020). For example, Lin et al. (2022) investigated the Chinese Mainland Mandarin and Vietnamese Mandarin Chinese using a synergetic-linguistic model incorporating the four dimensions of polysynonymy, polytextuality, frequency and word length. However, research on the holistic syntactic complexity of Mandarin Chinese varieties, particularly in relation to cognitive complexity in comprehension, remains limited.

A second limitation for data-driven research in overseas Mandarin Chinese variation is the difficulty of corpus construction and application. Existing resources, such as LIVAC² (Tsou & Kwong, 2015) and the Southeast Asian Chinese Media Corpus³ (Liu, 2020), lack accessible comprehensive word segmentation, POS tagging, and syntactic annotation. Although annotated corpora such as the Tagged Chinese Gigaword Corpus⁴ (Huang, 2009) are available, they primarily focus on Mandarin Chinese in the Chinese mainland and the Taiwan region, leaving Southeast Asian Mandarin without comparable resources.

Third, existing research mainly compares Mandarin Chinese used in the Chinese mainland with a single variety from outside the mainland, while comparative studies among Mandarin Chinese varieties spoken in various overseas communities remain limited. Political, cultural, and historical differences, along with linguistic contact environments, have shaped the evolutionary trajectories of Mandarin Chinese varieties. Some countries actively promote Mandarin Chinese, such as Singapore (Chin & Cavallaro, 2021), while others have historically imposed restrictive language policies that have significantly shaped the heritage and use of Mandarin Chinese, such as Indonesia (Setijadi, 2015). This issue is discussed in detail in [Section 3](#). Consequently, limiting research to comparisons between the Chinese Mainland Mandarin and its overseas variants offers only a partial perspective on the broader diversity of the language. Exploring the relationships among overseas Mandarin Chinese varieties offers a more comprehensive understanding of Mandarin Chinese variation and sheds light on how these varieties have adapted and evolved within diverse sociolinguistic contexts. In addition, existing research has paid insufficient attention to internal distinctions among outer circle countries where Mandarin Chinese is acquired as a heritage language.

2.3. Syntactic Complexity Metrics and Distributional Models

Sentence length has long been regarded as an important perspective in linguistic research (Köhler, 2012). Numerous studies have examined linguistic style using ASL or sentence length distributions, addressing issues such as genre variation (Chen & Kubát, 2024; Sichel, 1974; Yule, 1939), diachronic stylistic change (Savoy, 2017), and variation across World Englishes (Zhang & Liu, 2019). According to these studies, written registers typically feature longer and more complex sentences, whereas colloquial registers tend to exhibit shorter, less syntactically complex structures. Previous research has shown that sentence length distributions in human languages tend to exhibit consistent statistical regularities (Zhou et al., 2025). However, the specific distributional models may vary across languages. For instance, sentence length distributions in

English varieties have been successfully fitted to several statistical distributions, including the mixed negative binomial distribution, the Extended Positive and Negative Binomial (EPNB) distribution, and the negative binomial distribution (Liu et al., 2024; Zhang & Liu, 2019). Regarding Hindi, sentence length distributions are found to be better fitted by the EPNB model (Pande & Dhama, 2015). Similarly, sentence length distributions in Mandarin Chinese conform well to the EPNB distribution (Zhou & Liu, 2023, 2025). By contrast, sentence length distributions in Japanese are more accurately modelled using the Hyper-Pascal model (Ishida & Ishida, 2007).

Within dependency grammar, macro-level syntactic complexity is measured by dependency distance, the linear distance between a head and its dependent (Temperley, 2007). A greater MDD indicates higher syntactic complexity, reflecting increased cognitive load and heightened requirements for information processing (Liu, 2008). Consequently, MDD serves as a comprehensive indicator of syntactic cognitive complexity, making it useful for analysing linguistic variation (Chen & Kubát, 2024). Research shows that dependency distance distributions follow the right-truncated modified Zipf-Alekseev model (Jiang & Liu, 2015). Liu (2007) found that different languages exhibit varying dependency distances, with a cognitive threshold ensuring distances remain constrained (Liu, 2008). The right-truncated modified Zipf-Alekseev model parameters (a and b) have been shown to serve as sensitive indicators of variation in syntactic complexity (Hao et al., 2022). Zhang and Liu (2019) successfully fitted the dependency distance distributions of English varieties to the right-truncated modified Zipf-Alekseev model. Their analysis revealed that the parameters a and b positioned British English centrally, followed by Indian English, with other English varieties distributed around these two points. These findings suggest that the parameters a and b effectively differentiate the English varieties and provide empirical support for the three concentric circle theory of World Englishes proposed by Kachru (1985).

While these previous studies have demonstrated the effectiveness of ASL and MDD in measuring English linguistic variation, several limitations remain. First, existing research has mainly concentrated on inflectional languages such as English, with much less attention paid to the quantitative properties of isolating languages such as Mandarin Chinese. Second, there is a lack of large-scale empirical studies to determine whether the distributional characteristics of sentence length and dependency distance can effectively reflect variation among Mandarin Chinese varieties. Third, although previous research has shown that the right-truncated modified Zipf-Alekseev model parameters a and b are useful in differentiating English varieties, their applicability to Mandarin Chinese varieties has yet to be verified.

3. Current Research

Inspired by Zhang and Liu (2019), this study employs quantitative methods to advance data-driven research on Mandarin variation. Accordingly, we constructed a syntactically annotated corpus⁵ comprising over one million words, including two overseas Mandarin Chinese varieties (SM and IM) and one reference corpus of Chinese Mainland Mandarin (CMM). SM and IM constitute prototypical cases of stable and disrupted trajectories of overseas Mandarin Chinese heritage respectively, and the sociolinguistic ecologies in which they are embedded stand in sharp contrast. In Singapore, Mandarin Chinese is an official language with strong governmental support, reinforced through education, media, and policies like the 1979 *Speak Mandarin Campaign*, ensuring its stable transmission (Bokhorst-Heng & Wee, 2007; Chua, 2010). In contrast, Indonesia experienced over 30 years of restrictive policies under Suharto, banning Chinese education and publications, leading to weakened linguistic heritage identity among Chinese Indonesians (Hoon, 2006; Sari et al., 2018). Following the resignation of Suharto in 1998, the revival of Mandarin Chinese in Indonesia continues to face difficulties and obstacles (Harjatanaya & Hoon, 2020; Setijadi, 2015). We hypothesize that these differing language policies and contact environments have contributed to certain differences in the syntactic characters of SM and IM.

This study tests the hypothesis by using ASL and MDD to compare the macro-level syntactic complexity of two overseas varieties of Mandarin Chinese with that of CMM. Following Zhou and Liu (2023, 2025), Chinese sentence length distributions were fitted using the EPNB model, while dependency distance distributions were modelled by the right-truncated modified Zipf-Alekseev distribution. Specifically, it focuses on addressing the following three research questions:

- (1) Do the three Mandarin Chinese varieties differ in ASL? Do their sentence length distributions follow the EPNB distribution? Can the parameters of EPNB model distinguish among the three varieties?
- (2) Do the three Mandarin Chinese varieties differ in MDD? Do their dependency distance distributions follow the right-truncated modified Zipf-Alekseev distribution? Can the parameters of the right-truncated modified Zipf-Alekseev model distinguish among the three varieties?
- (3) What sociolinguistic and contextual factors may account for the variation in ASL and MDD across the three Mandarin Chinese varieties?

4. Methods and Corpus

4.1. Corpus Selection

Hou and Huang (2020) note that regional variation often interacts with linguistic features associated with genre in the automatic classification of Mandarin Chinese varieties, resulting in complex patterns of variation. This makes it difficult to determine whether an observed difference arises from genre or from regional factors. The present study focuses on analysing the editorial genre across different Mandarin Chinese varieties. Editorials are a subgenre of the journalistic genre, intended to present opinions on topical issues in order to persuade readers and disseminate institutional ideologies (Connor, 1996). Compared with news reports, which emphasize neutrality and factual accuracy, editorials are typically authored collectively by an editorial board and published anonymously (Ciofalo, 1998), reflecting an institutional voice and characterized by a high degree of impersonality (Firmstone, 2008). They are intended to articulate the stance of the newspaper, sometimes even representing the voice of the government, guiding public opinion, and influencing social cognition (Farrokhi & Nazemi, 2015). In addition, editorials have for many years been a valuable corpus for comparing languages across cultures and an important focus in studies of World Englishes. A fruitful body of research has already appeared and has produced important insights (Ansary & Babaii, 2009; Bonyadi, 2010; Huang & Ren, 2020; Zarza & Tan, 2016). For example, Huang and Ren (2020) found that China Daily editorials tend to employ a linguistic style that is predominantly informational and impersonal, whereas The New York Times editorials are characterized by a more narrative and argumentative style. Taking the above into consideration, selecting the editorial genre as the focus of the present study is both methodologically sound and analytically productive.

The CMM corpus includes editorials from *People's Daily*, *Guangming Daily*, and *the Chinese-language Global Times*. The SM corpus is based on editorials from *Lianhe Zaobao*. The IM corpus consists of editorials from *Harian Indonesia*, *Qiandao Ribao*, and *Shang Bao*. Editorial texts from 2013 to 2024 were automatically collected using Python scripts, with procedures to ensure randomness and balance, followed by manual cleaning to remove reprinted editorials originating from the Chinese mainland. The final corpus exceeds one million words, with basic statistics in Table 1.

Table 1. Overall corpus statistics information.

Mandarin Chinese varieties	Number of Editorials	Total Tokens	Total Sentences
CMM	1,100	727,719	27,907
IM	1,273	724,160	34,546
SM	1,021	719,864	28,521

4.2. Newspaper Background

People's Daily, *Guangming Daily*, and the *Chinese-language Global Times* are national media outlets in the Chinese mainland and are characterized by a strong official orientation. *Lianhe Zaobao* is the highest-circulation Chinese-language newspaper in Singapore and one of the most representative Chinese-language media outlets in Southeast Asia.⁶ *Harian Indonesia*, *Qiandao Ribao*, and *Shang Bao* are all major Chinese-language newspapers in Indonesia. *Qiandao Ribao* maintains a stable distribution network in eastern Indonesia and its readership mainly comprises Chinese Indonesians⁷. *Harian Indonesia* operates printing and distribution hubs in major cities such as Jakarta, Pontianak, Surabaya, and Medan, and it commands a substantial Chinese-Indonesian readership throughout Indonesia and across Southeast Asia.⁸ *Shang Bao*, founded in 1998 during the era of democratic reform in Indonesia, was the first Mandarin Chinese newspaper of that period and the first daily devoted primarily to financial news. Its readership consists chiefly of Chinese Indonesian entrepreneurs and other business professionals interested in economic policy.⁹ According to the official website of the Ministry of Foreign Affairs of the People's Republic of China, all four Southeast Asian newspapers selected for this study are recognized as representative Chinese-language newspapers in Singapore¹⁰ and Indonesia.¹¹

During the data collection period, all newspapers selected for this study maintained editorial columns and catered to the mainstream local Chinese-speaking readership. In terms of media influence, each occupies a prominent position among the most representative Chinese-language outlets in its respective country, characterized by broad coverage and pronounced agenda-setting capacity on general issues and public opinion. Given their comparability in genre, audience orientation, and functional roles, these newspapers provide a solid basis for comparative analysis in this study.

4.3. Corpus Processing and Data Collection

A syntactically annotated corpus based on dependency grammar provides effective support for large-scale quantitative syntactic studies. Syntactic dependency relations are typically defined by three core properties that are widely acknowledged in linguistic research (Hudson, 2007; Liu, 2009). First, they are binary, involving exactly two linguistic units. Second, they are generally asymmetrical and directed, with one unit functioning as the head and the other as the dependent. Third, dependency relations are labelled, meaning that the type of syntactic connection is explicitly specified, often through a label placed above the arc linking the two elements in syntactic representations. Identifying these dependency relations is an essential task in

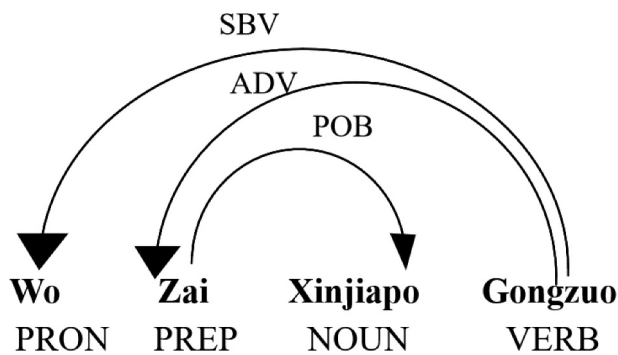


Figure 1. The dependency structure of the example sentence *I work in Singapore*.

syntactic analysis (Hudson, 2007). Based on the properties of dependency grammar, a dependency structure can be represented by directed arcs. Figure 1 illustrates the dependency analysis of the sentence ‘*Wǒ zài xīnjiāpō gōngzuò*’ (I work in Singapore), where the syntactic labels on the arcs indicate the specific types of dependency relationships between linguistic units.

This study employed the Language Technology Platform (LTP)¹² (Che et al., 2021) to automatically perform part-of-speech tagging and dependency annotation of the corpus. Following annotation, we conducted manual proofreading. Subsequently, we used Python programming to organize dependency information for each sentence into the format illustrated in Table 2.

For any dependency relation involving the words W_a (the head) and W_b (the dependent), where a and b represent the word orders of W_a and W_b , the dependency distance is defined as the absolute difference between a and b . Since the headword may either precede or follow the dependent word, the dependency distance value can be positive or negative. The absolute value of dependency distance is used to calculate the MDD for a sentence or a treebank. Since a sentence with n words always contains $n-1$ dependency relations, the dependency

Table 2. Dependency structure of the example sentence ‘*wǒ zài xīn jiāpō gōng zuò*’ (I work in Singapore).

Dependent			Head			Dependency relation	Absolute dependency distance
word order	word	POS	word order	word	POS		
1	wǒ	r	4	gōng zuò	v	SBV	3
2	zài	p	4	gōng zuò	v	ADV	2
3	xīn jiāpō	n	2	zài	p	POB	1
4	gōng zuò	v	0	/	/	HED	0

distance of a sentence containing n words can be determined using Equation 1.

$$MDD(\text{The sentence}) = \frac{1}{n-1} \sum_{i=1}^{n-1} |DD_i| \quad (1)$$

In a sentence, the only word without a head is the root node of the syntactic tree, which is labelled as HED. The dependency distance of HED is considered to be zero. By subtracting the word order number of the head from the word order number of its dependent word, we can obtain several dependency distance values for the example sentence: 3, 2, -1, and 0. Using formula (1), the MDD of the sample sentence is calculated as $(|3| + |2| + |-1|)/3 = 2$. If formula (1) is extended to an entire treebank, the MDD of the entire treebank can also be determined. If a treebank contains m words and n sentences, then the MDD of the treebank is calculated by Equation 2:

$$MDD(\text{The treebank}) = \frac{1}{m-n} \sum_{i=1}^{m-n} |DD_i| \quad (2)$$

In Equation 1 and Equation 2, DD_i represents the dependency distance of the i th dependency relation in the sentence or the treebank.

This study employs a right-truncated modified Zipf-Alekseev model to fit the dependency distances, with the fitting Equation 3 given as follows:

$$P_x = \begin{cases} \alpha, & x = 1 \\ \frac{(1-\alpha)x^{(a+b \ln x)}}{T}, & x = 2, 3, \dots, (n) \end{cases} \quad (3)$$

Where

$$T = \sum_{j=2}^n j^{-(a+b \ln j)}, \quad a, b \in R, \quad 0 < \alpha < 1$$

The ASL for different Mandarin Chinese varieties was also calculated. ASL has long been regarded as a measure of syntactic complexity, and higher MDD is often closely associated with longer sentences (Ferrer-I-Cancho, 2014). In this study, periods, question marks, and exclamation marks were used as delimiters to calculate the length and number of complete sentences.¹³ The method for calculating ASL is provided in Equation 4.

$$ASL = \frac{\text{Total Number of Words}}{\text{Total Number of Sentences}} \quad (4)$$

Research indicates that sentence length distributions in language conform to the Extended Positive Negative Binomial (Pande & Dhami, 2015; Zhou & Liu, 2025). This distribution model is shown in Equation 5.

$$P_x = \begin{cases} 1 - \alpha, & x - 1 = 0 \\ \alpha \binom{k+x-2}{x-1} p^k q^{x-1}, & x - 1 = 1, 2, 3, \dots, n-1 \end{cases}; \quad \begin{matrix} k > 0, 0 < p, \alpha < 1 \\ q = 1 - p \end{matrix} \quad (5)$$

The sentence length, dependency distance, and distribution data in this study were all generated using Python scripts, and the software Altmann-Fitter (2013) was used for data fitting.

5. Results

5.1. Average Sentence Length and Sentence Length Distribution Across Mandarin Chinese Variants

As shown in Figure 2, the ASL for CMM, IM and SM are 26.08, 20.96 and 25.24 respectively. Using sentence length as the dependent variable and variety as the independent variable, a Kruskal-Wallis test¹⁴ reveals statistically significant differences among the three groups ($H(2) = 649.27$, $p < 0.001$, $\eta^2 = 0.191$). Subsequent Bonferroni-corrected post hoc tests show that the ASL of CMM is significantly longer than those of IM and SM ($ps < 0.001$), while IM has significantly shorter sentence lengths compared to the other two groups ($ps < 0.001$). The ASL of SM is positioned between those of CMM and IM, showing a statistically significant difference from both ($ps < 0.001$).

Figure 3 illustrates the sentence length distributions of CMM, IM and SM. In terms of overall distributional patterns, all three Mandarin Chinese

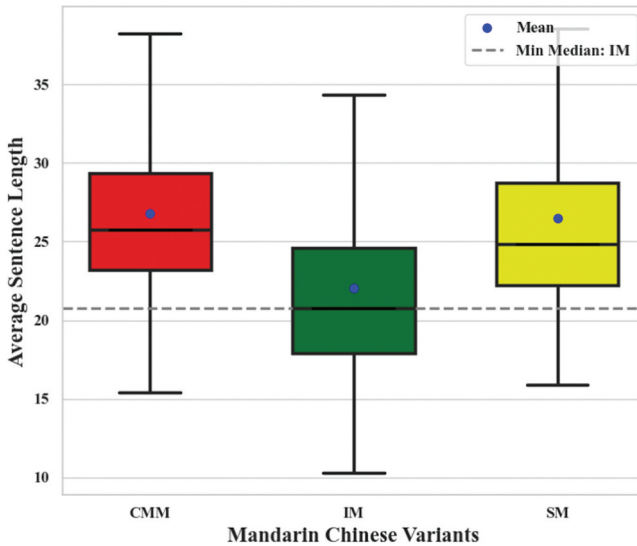


Figure 2. Boxplot of average sentence length of three Mandarin Chinese varieties.

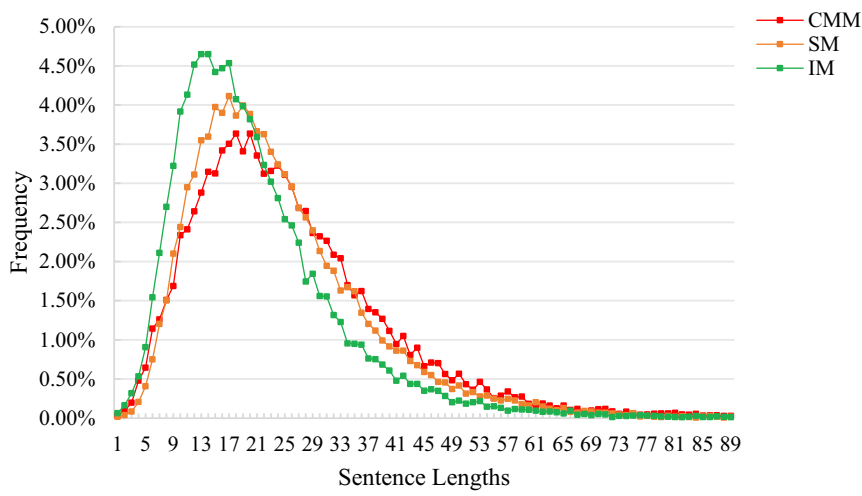


Figure 3. Sentence length frequency distribution of three Mandarin Chinese varieties.

varieties exhibit a right-skewed distribution, indicating a significant predominance of shorter sentences.

We used Altmann-Fitter to fit the sentence length distributions of the three Mandarin Chinese varieties to the EPNB distribution. The results are presented in Table 3. Based on the R^2 values, the sentence length distributions of all three varieties were successfully fitted to the EPNB distribution ($R^2 > 0.9$). This distribution pattern aligns with the principle of least effort (Zipf, 1949). The sentence length distribution curves for CMM, SM, and IM reveal a consistent pattern in which sentences of moderate length dominate across all three varieties, while extremely long sentences occur only rarely. This tendency is not only observed in Mandarin Chinese varieties but also aligns with findings on English varieties, where the ASL fluctuates within a relatively narrow band (24.5 to 29.2 words) (Zhang & Liu, 2019), indicating that cross-linguistic consistency may exist in sentence length distributions across different language.

Although the sentence length distributions for all three Mandarin Chinese varieties fit well to the EPNB Model, the differences in parameter values indicate variations in the specific characteristics of each Mandarin Chinese

Table 3. Fitting ASL of three Mandarin Chinese varieties to the EPNB model.

Mandarin Chinese varieties	Average Sentence Lengths	<i>k</i>	<i>p</i>	<i>a</i>	<i>C</i>	<i>R</i> ²
CMM	26.077	3.4041	0.1267	0.9992	0.0604	0.9500
SM	25.240	3.5872	0.1437	0.9986	0.1662	0.9547
IM	20.962	3.7848	0.1667	0.9983	0.0475	0.9631

**k*, *p*, *a*-Model parameters; *C*, *R*²-Fitting parameters.

variety. The EPNB model is derived from the negative binomial distribution (Tuzzi et al., 2012; Liu et al., 2024). The parameter α is determined by the probability of the shortest sentence and reflects the principle of least effort, as smaller values of α are associated with a higher likelihood of the shortest sentence occurring. The parameters k and p act as tuning coefficients that determine the distribution by modulating its kurtosis and skewness, thereby capturing the relative prevalence of long and short sentences (Zhou & Liu, 2025). Overall, k , p , and α constitute complementary quantitative indicators of the principle of least effort. The Kruskal-Wallis test revealed significant differences among the three Mandarin Chinese varieties in both parameters p and α ($H_p(2) = 11.043$, $p = .004$; $H_\alpha(2) = 12.760$, $p = .002$). Taking α as an example, IM exhibits a lower α value (0.9983) than CMM (0.9992) and SM (0.9986), indicating a higher proportion of short sentences in its sentence length distribution. This result is consistent with Figure 3, which shows the IM distribution peaking furthest to the left (around 13 words) and reaching the highest peak (exceeding 4.5%), indicating its pronounced concentration in the short sentence range. In comparison, SM shows a more balanced sentence length distribution, with proportions of short and long sentences that are moderate, a peak lower than that of IM but higher than that of CMM. CMM demonstrates the broadest sentence length range, with pronounced fluctuations in the frequency of long sentences (above 40 words). Its distribution exhibits the lowest peak and the most distinct long-tail effect, characterized by strong tail extensibility.

While sentence length distribution reflects some features in the information expression preferences of different Mandarin Chinese varieties, it alone is insufficient for a comprehensive assessment of syntactic complexity. This study further examines MDD in CMM, IM, and SM to explore variations in dependency syntactic complexity.

5.2. Mean Dependency Distance and Dependency Distance Distribution Across Mandarin Chinese Variants

As shown in Figure 4, variations in MDD for mixed sentence lengths are observed across different Mandarin Chinese varieties. CMM exhibits the highest MDD at 3.26, followed by SM at 3.12, while IM records the lowest MDD at 2.99. We conducted a Kruskal-Wallis test for significance, with dependency distance from the three original datasets as the dependent variable and variety as the independent variable. The results indicate significant differences in MDD across the three Mandarin Chinese varieties ($H(2) = 339.425$, $p < 0.001$, $\eta^2 = 0.099$). A Bonferroni-corrected post hoc test further reveals that CMM exhibits a significantly higher MDD than both IM and SM ($ps < 0.001$), while IM has a significantly lower MDD than the other two varieties ($ps < 0.001$). The MDD of SM is intermediate between CMM

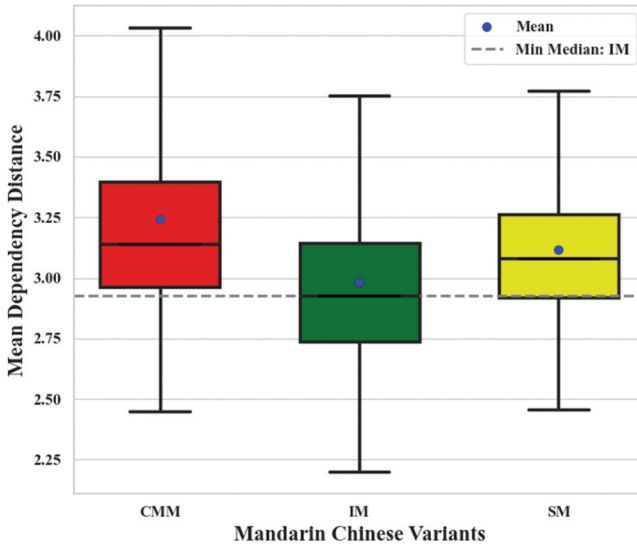


Figure 4. Boxplot of dependency distance of three Mandarin Chinese varieties.

and IM but remains significantly different from both ($ps < 0.001$). These findings suggest that CMM exhibits greater syntactic complexity and processing difficulty than SM and IM, while SM demonstrates higher complexity than IM. The results align with those obtained by Zhang and Liu (2019) in their study on English varieties, further reinforcing the role of dependency distance as an effective metric for assessing syntactic complexity and a crucial variable in understanding language variation across different sociocultural contexts.

Previous research has demonstrated that MDD is positively correlated with sentence length, meaning that longer sentences tend to have higher MDD values (Ferrer-I-Cancho & Liu, 2014; Jiang & Liu, 2015). In order to better interpret the observed differences in MDD across Mandarin Chinese varieties, this section explores the interaction between sentence length and dependency distance. Following Liu et al. (2025), sentence length was divided into 10 intervals: [0–10], [11–20], [21–30], [31–40], [41–50], [51–60], [61–70], [71–80], [81–90], and [90+]. The MDD for each interval is shown in Figure 5. Pearson correlation analyses revealed a strong and statistically significant positive relationship between sentence length and MDD across all three Mandarin Chinese varieties. CMM showed the highest correlation ($r = 0.988$, $p < 0.001$), followed by SM ($r = 0.985$, $p < 0.001$) and IM ($r = 0.983$, $p < 0.001$). These results confirm that longer sentences are associated with increased dependency distance, while also highlighting differences in how the three varieties structure syntactic complexity. CMM consistently exhibits the highest MDD across longer sentence intervals

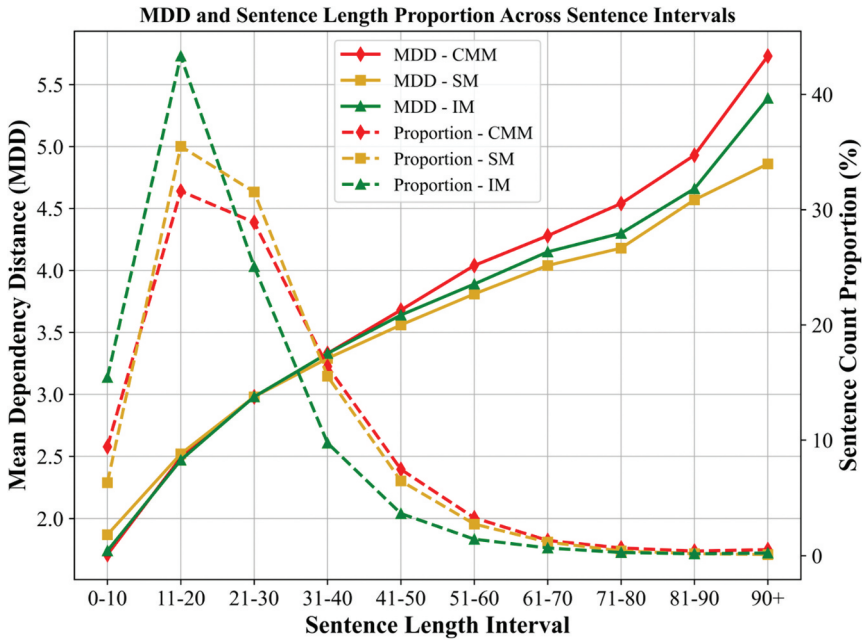


Figure 5. Mean dependency distance across different sentence length intervals of three Mandarin Chinese varieties.

(starting from the 31–40-word interval). This is accompanied by a wide range of sentence lengths with a prominent long tail pattern, particularly beyond the 40-word threshold, indicating greater syntactic complexity and flexibility. SM presents a more balanced sentence length distribution, with slightly elevated MDD values in the shortest interval (0–10 words) and a relatively moderate rate of increase in longer sentences, suggesting a more controlled approach to syntactic expansion. In contrast, IM exhibits a strong preference for shorter sentences, with 58.77% of its sentences falling within the 0–20 word range. This concentration of shorter sentences is associated with the lowest overall MDD among the three varieties. Although IM shows increasing MDD values in longer sentence intervals, the limited proportion of long sentences leads to a generally lower level of syntactic complexity compared to CMM and SM.

Human languages exhibit a universal tendency towards Dependency Distance Minimization (Futrell et al., 2015; Liu, 2008). The dependency distance between a head word and its dependent in a sentence can not expand indefinitely. This principle of least effort leads to the dependency distance and its frequency distribution in languages following a power-law distribution, which can be fitted by the right-truncated modified Zipf-Alekseev distribution with the notable advantage of being independent of

sentence length (Jiang & Liu, 2015). Zhang and Liu (2019) found that the parameters of the right-truncated modified Zipf-Alekseev distribution can effectively distinguish English variants. Building upon previous research findings, this section further examines the distribution characteristics of dependency distance across various Mandarin Chinese variants within the editorial genre. Figure 6 presents the probability distribution of dependency distances for CMM, IM, and SM, with all three varieties demonstrating the long-tail pattern characteristic of power-law distributions.

Table 4 provides a summary of the fitting results for the dependency distance distributions of the three Mandarin Chinese varieties using the right-truncated modified Zipf-Alekseev model. The model demonstrates a highly satisfactory fit across all varieties, with R^2 values exceeding 0.99 and C all below 0.02, indicating an excellent overall fit. Although $P(X^2)$ for all varieties are reported as 0.000, this is primarily attributed to the large sample size involved (Mačutek & Wimmer, 2013). As noted in previous studies, the chi-square test tends to reject the null hypothesis in large datasets regardless of actual model fit quality. Therefore, we did not rely solely on $P(X^2)$ as the criterion. Instead, we placed greater emphasis on C and R^2 , which are recognized as more stable and reliable measures for assessing model fit in large-scale data (Guan, 2019; Hao et al., 2022). These two metrics consistently confirmed the suitability of the modified Zipf-Alekseev model for representing the dependency distance distributions across all three Mandarin Chinese varieties.

The results suggest a consistent trend of dependency distance minimization across the three Mandarin Chinese variants. This implies that speakers of different Mandarin Chinese varieties universally exhibit a tendency to construct sentences with shorter dependency distances and consequently lower cognitive load. Such a phenomenon is consistent with the universal principles and patterns of human language and aligns with previous findings on English variants (Zhang & Liu, 2019). The trend towards dependency distance minimization further reveals that speakers of Mandarin Chinese varieties, despite diverse backgrounds, employ similar cognitive strategies in sentence construction.

We find that the parameters a and b of the right-truncated modified Zipf-Alekseev model show differences across Mandarin Chinese variants. For parameter a , CMM exhibits the highest value (0.5529), followed by SM (0.3366), and then IM (0.3352). Conversely, for parameter b , IM displays the highest value (0.4826), followed by SM (0.4689), while CMM has the lowest value (0.3925). Although the Kruskal-Wallis test showed no significant group difference in parameter a ($p = .128$), Spearman's rank correlation revealed a significant negative association for this parameter ($\rho = -.341$, $p = .024$, 95%[-.577, -.053]). Parameter b differed significantly among the three Mandarin Chinese varieties in a Kruskal-Wallis test ($H(2) = 16.214$,

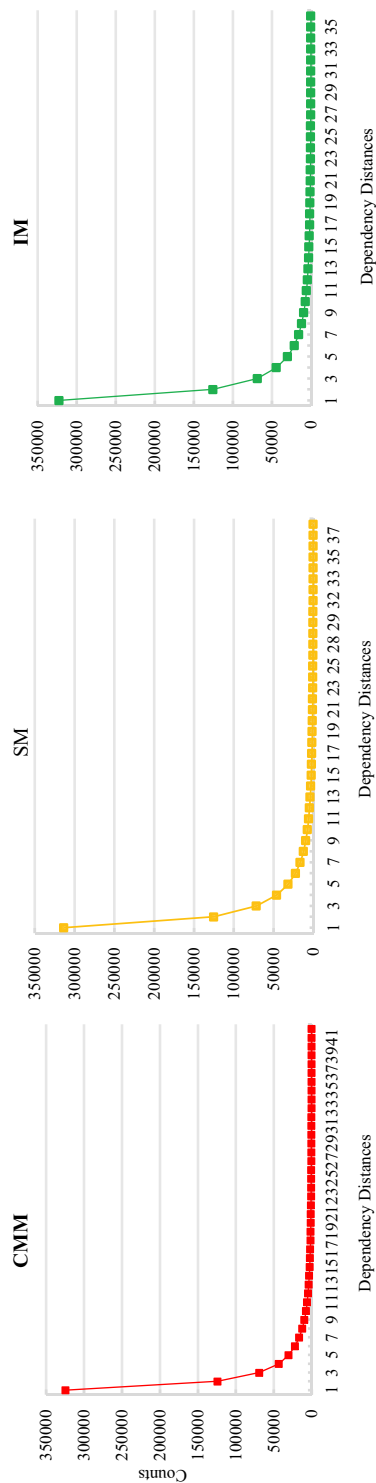


Figure 6. Frequency distribution of dependency distances of three Mandarin Chinese varieties.

Table 4. Fitting results of the right truncated modified Zipf-Alekseev distribution for dependency distances of different Mandarin Chinese varieties.

Mandarin Chinese varieties	<i>a</i>	<i>b</i>	χ^2	$P(\chi^2)$	DF	<i>C</i>	R^2
CMM	0.5529	0.3925	165.3645	0.0440	82.4667	0.0036	0.9993
SM	0.3366	0.4689	97.7343	0.0501	66.2667	0.0022	0.9996
IM	0.3352	0.4826	140.5117	0.0151	69.7333	0.0031	0.9995

**a*, *b*-the parameters of the right truncated modified Zipf-Alekseev model; χ^2 -Chi-square; *DF*-degrees of freedom; *p*- probability of chi-square; *C*-discrepancy coefficient; R^2 -goodness of fit.

$p < .001$, $\eta^2 = .376$). These results align with the earlier findings regarding the ranking of MDD across the three Mandarin Chinese varieties: the parameter values for the dependency distance fitting of SM, which benefits from relatively favourable Chinese education and policies, are closer to those of CMM. By comparison, the parameter values for the dependency distance fitting of IM, which has a historical background of interrupted Chinese language education, are further from those of CMM. To further quantify the similarity among the three Mandarin Chinese varieties in terms of dependency distance, we utilize the parameters *a* and *b* from the right-truncated modified Zipf-Alekseev distribution as coordinates and calculate their pairwise Euclidean distances. The results demonstrate that SM shows a closer alignment with CMM, with a Euclidean distance of 0.0214, whereas IM exhibits a greater Euclidean distance of 0.0303 relative to CMM. This indicates that the dependency distance distribution of SM is more similar to that of CMM. The findings of this study align closely with those of Zhang and Liu (2019), which observed that the fitting parameters of different English variants fluctuate around those of British and Indian English, where English is either a native or official language. This demonstrates that the parameters *a* and *b* from the probability distribution fitting of dependency distances are not only applicable for distinguishing English variants but also suitable for isolating Mandarin Chinese varieties, highlighting their potential universal typological applicability.

6. Discussion

The findings presented in Section 5 reveal significant differences in sentence length and dependency distance across the three Mandarin Chinese varieties. While quantitative metrics such as ASL, MDD, and model parameters provide a foundation for identifying observable contrasts, a more comprehensive understanding of these variations necessitates consideration of underlying linguistic structures, historical transmission, sociolinguistic contexts, and patterns of language contact.

6.1. Explanations for Sentence Length Differences

Longer sentences tend to include not only more words but also more complex syntactic structures, which demand greater cognitive effort and processing time across reading, writing, and pronunciation, making them inherently more difficult to produce and understand (Sigurd et al., 2004). As a result, longer sentences are typically associated with a higher level of textual formality, serving as an effective means of conveying complex ideas and structured logical reasoning in formal discourse, whereas shorter sentences are more commonly found in informal contexts, such as spoken communication, where the ease of real-time cognitive processing is more important (Wang & Liu, 2017; Liu et al., 2017). Therefore, the tendency for IM to exhibit shorter sentence lengths may reflect a greater tendency towards less formal and more colloquial language use. This is in line with a recent large-scale sociolinguistic survey indicating that IM is predominantly used for informal communication within families and communities (Yuan, 2023). This context, coupled with the historical interruption of Chinese education in Indonesia, may explain why IM exhibits a stronger tendency towards colloquial features.

The balanced sentence length distribution in SM reflects the multilingual environment of Singapore, which is characterized by the coexistence of multiple official languages. As one of the official languages in Singapore, Mandarin Chinese has reached a dynamic balance between formality and informality. On the one hand, government support for Mandarin Chinese has promoted the standardization of its formal written discourse. Since the launch of the *Speak Mandarin Campaign* in 1979, the Singaporean government has officially aimed to shift speakers of various Chinese dialects towards Mandarin Chinese (Bokhorst-Heng & Wee, 2007). The language standardization policy driven by the Singapore government has promoted greater formalization in Mandarin Chinese, ensuring a consistent level of linguistic formality. On the other hand, the multilingual setting of Singapore facilitates linguistic interaction, leading to language simplification and enhanced communicative efficiency (Trudgill, 2011). As a result, SM employs economical expressions while maintaining the necessary degree of formality. This balance between linguistic standardization and communicative efficiency contributes to the relatively balanced distribution of sentence length in SM. By contrast, The sentence length distribution in CMM exhibits a pronounced formal characteristic. The promotion of Mandarin Chinese in the Chinese Mainland has long been shaped by official language policies focused on standardization. Since 1956, the Chinese government has systematically promoted Mandarin as the national lingua franca, integrating it extensively into education, media, government documents, and other formal domains (Feng & Adamson, 2019). This policy has driven the progressive

standardization of written Mandarin, leading to highly regulated linguistic norms and the gradual development of more complex syntactic structures to meet the expressive demands of formal registers, such as editorials.

6.2. Explanations for Dependency Distance Differences

This section examines four explanatory pathways, namely conservatism, convergence, Chinese dialectal substratum, and simplification, each offering a distinct yet complementary perspective on how the dependency distance of SM and IM have varied from those of CMM. While overall MDD serves as a macro-level measure of syntactic complexity, its variation is shaped by the relative usage of particular dependency types. To better interpret the observed differences in MDD across these Mandarin Chinese varieties, we consider not only macro-level factors such as language policy and socio-linguistic context but also the distribution patterns of specific dependency relations. To clarify the variation in dependency relations, Table 5 presents the proportions of each dependency type across the three Mandarin Chinese varieties. Accordingly, this section integrates a micro-level analysis of selected dependency relations that are especially relevant to understanding structural variation. A brief description of each dependency label, including its syntactic function and examples, is provided in Appendix A (Table A1) to facilitate interpretation.

6.2.1. Conservatism as an Explanation

From a holistic perspective, Liu (2008) posits that the existence of the shortest path between any two heads in a syntactic network is associated with the minimization of dependency distance, and that MDD and mean path length exhibit certain similarities (Hao et al., 2021). From a diachronic perspective, the syntactic complexity of Chinese has generally increased over

Table 5. Proportional distribution of 12 major dependency relation types across three Mandarin Chinese varieties.

The ratio of dependency types	CMM	SM	IM
ATT_ratio**	27.45%	27.41%	27.82%
ADV_ratio**	18.60%	19.34%	18.08%
VOB_ratio**	12.97%	12.99%	13.38%
COO_ratio**	10.51%	8.85%	9.38%
RAD_ratio**	8.83%	8.39%	7.76%
SBV_ratio**	8.81%	9.30%	9.31%
POB_ratio**	4.47%	4.78%	4.64%
HED_ratio**	3.86%	3.97%	4.78%
LAD_ratio**	1.59%	2.39%	2.13%
CMP_ratio**	1.48%	1.15%	1.25%
FOB_ratio**	0.77%	0.60%	0.64%
DBL_ratio**	0.63%	0.80%	0.80%

Note: Significance codes for the Kruskal-Wallis test: ** $p < 0.01$.

the centuries. Chen et al. (2018) found that the mean path length in Chinese has risen over time, suggesting that dependency distance increases in more modern stages of Chinese and decreases in earlier ones. Existing research has shown that immigrant languages tend to be more conservative than their counterparts in the homeland, generally exhibiting a slower rate of language change (Haugen, 1938; Montrul, 2016). This phenomenon is substantiated by the findings in the current study.

The linguistic divergence between overseas Mandarin Chinese and its counterpart in the Chinese Mainland began during the early 20th century, particularly following the May Fourth Movement and the Vernacular Chinese Movement (Chen, 1999, p. 97). Under the influence of official language promotion and a standardized educational framework, the syntactic structure of CMM has undergone systematic standardization and reinforcement. Simultaneously, overseas Mandarin Chinese varieties have evolved distinct linguistic trajectories due to the diverse language environments in which they exist. While all three Mandarin Chinese varieties differ significantly in statistical terms, SM (MDD = 3.12) shows a relatively smaller deviation from CMM (MDD = 3.26) in terms of mean value, whereas IM (MDD = 2.99) departs the most from CMM in terms of dependency distance. These syntactic differences can be partly attributed to the historical language policies adopted in different countries. Singapore, as one of the earliest countries where Mandarin Chinese was widely adopted, has a highly concentrated Chinese-speaking population. The relatively stable transmission of Mandarin Chinese has been reinforced by the combined influence of Chinese education, media, and literature. Additionally, the launch of the *Speak Mandarin Campaign* by the Singaporean government in 1979 further accelerated the promotion of standardized Mandarin Chinese (Chin & Cavallaro, 2021), bringing SM into closer alignment with the syntactic developments observed in modern Mandarin Chinese. In contrast, the linguistic policy landscape in Indonesia has followed a markedly different trajectory. The historical anti-Chinese movements and assimilation policies severely disrupted the transmission of Chinese and Chinese cultural heritage. The prolonged absence of a structured Chinese education system within the Indonesian Chinese community has resulted in linguistic discontinuities, limiting the natural development and innovation of IM. Consequently, IM exhibits more conservative linguistic characteristics, suggesting a relatively slower pace of change.

6.2.2. *Convergence as an Explanation*

Research has shown that languages undergoing prolonged contact with multiple linguistic systems often undergo convergence and gradual localization (Thomason & Kaufman, 2001). SM and IM serve as representative cases of such linguistic contact, as Mandarin Chinese in these countries has been influenced by external languages such as English and Indonesian over an

extended period. This process of language contact and evolution has contributed to the localization of SM and IM at the syntactic level, leading to differences in syntactic complexity compared to CMM. For instance, IM exhibits the lowest RAD proportion (7.76%) and the highest ATT proportion (27.81%) among the three Mandarin Chinese varieties. The Kruskal-Wallis test results indicate significant differences in the use of these two dependency relations across the three varieties ($H_{\text{ATT_ratio}(2)} = 8.529, p < 0.01, \eta^2 = .003$; $H_{\text{RAD_ratio}(2)} = 272.15, p < 0.001, \eta^2 = .077$). In Mandarin Chinese, RAD is predominantly composed of phrases containing the structural particle *de* (of), which functions as a syntactic link between attributive modifiers and head nouns. Statistical analysis shows that *de* (of) is the most frequently occurring RAD component across all three varieties. However, IM exhibits the lowest absolute frequency of RAD structures formed by *de* (of) among the three Mandarin Chinese varieties. This may suggest that IM exhibits a greater tendency for direct linking between attributive modifiers and head nouns, with less reliance on the explicit marking of modification through the structural particle *de* 'of'. In contrast to Chinese, which commonly employs *de* (of) to mark adjectival modification, Indonesian frequently omits the attributive marker *yang*, using word order alone to convey the modifying relationship (Ewing, 2013; Moeliono et al., 2017). Therefore, we hypothesize that IM, through prolonged contact with Indonesian, has gradually reduced its reliance on the structural particle *de* (of), resulting in a tighter modifier-head relationship.

Similarly to the influence of Indonesian on IM, SM has been largely influenced by English. Wong (2024) points out that due to prolonged exposure to a bilingual environment, SM has gradually converged with English in terms of semantic expression, speech habits, and cultural values. This study further corroborates this pattern, as evidenced by the increased use of prepositional-object (POB) and adverbial-modifier (ADV) structures, reflecting the impact of English on SM at the syntactic level. Specifically, SM exhibits the highest proportion of POB constructions (4.78%), exceeding IM (4.64%) and CMM (4.47%). The Kruskal-Wallis test results indicate that the differences in POB usage among the three Mandarin Chinese varieties are statistically significant ($H(2) = 47.286, p < 0.001, \eta^2 = .013$). Chinese and English employ different strategies for sentence expansion. From a syntactic perspective, attributive dependency is the most frequently used syntactic feature in Chinese, primarily because Chinese places less emphasis on grammatical mechanisms for post-nominal modification compared to English (Qin & Zhou, 2019). In contrast, prepositional-object structures constitute one of the most prominent syntactic constructions in English. The English prepositional system is more extensive and functionally diverse than that of Chinese, making prepositions one of the most frequently utilized functional categories in English sentence construction. English prepositional

structures serve a wide range of grammatical functions and are extensively used to encode temporal, locative and directional relations (Huddleston & Pullum, 2005). Thus, the prevalence of POB constructions in SM supports the view that extended contact with English has had a marked effect on the development of the variety. Additionally, SM exhibits the highest frequency of ADV (19.34%), significantly exceeding that of CMM (18.60%) and IM (18.08%). Statistical analysis further confirms that the differences in ADV frequency across the three Mandarin Chinese varieties are statistically significant ($H(2) = 60.369$, $p < 0.001$, $\eta^2 = .021$). In Chinese, prepositional phrases, composed of a preposition and its object, function syntactically as adverbials and establish a close relationship with the predicate. The influence of English syntactic features on SM has resulted in a structural tendency to use prepositional phrases as adverbials more frequently than in CMM. Despite English contact leading to more frequent use of Chinese prepositions, the overall dependency distance in SM remains lower than that in CMM. We hypothesize that SM may reduce the complexity of attributive modifiers during syntactic expansion, while adjusting the expression of syntactic information through adverbial-central structures and preposition-object constructions. This hypothesis is further supported by the fact that SM exhibits the lowest ATT proportion among the three Mandarin Chinese varieties (27.4%).

6.2.3. *Dialect Substratum as an Explanation*

In addition to the influence of foreign languages, overseas Mandarin Chinese has retained linguistic features from Southern Chinese dialects such as Hokkien, Hakka, and Cantonese (Bradley, 1992; Lee, 2003; Lin & Khoo, 2018). For example, the Kruskal-Wallis test reveals significant differences in the frequency of SBV structures among the three Chinese variants ($H(2) = 30.997$, $p < 0.001$). Bonferroni-corrected post hoc analyses indicate that SM with 9.30% and IM with 9.31% exhibit higher SBV frequencies compared to CMM at 8.81% ($ps < 0.001$). However, no significant difference is observed between SM and IM ($p = .325$), indicating that these two variants appear to share similar patterns in the use of subject-predicate structures. We posit that this phenomenon may correlate with the variation in the acceptability of topic-comment structures between Mandarin Chinese and Southern Chinese dialects. As a linguistic system grounded in Northern Mandarin dialects, Mandarin Chinese exhibits greater tolerance for topic prominence and null subjects (Li & Thompson, 1976), illustrated by the prevalence of topic subjects and subject-drop sentence structures. In comparison, Chinese dialects such as Cantonese and Southern Min impose more restrictive conditions on the acceptability of topic-comment structures and null subjects (Paul, 2002; Zhang & Deng 2011; Yiu, 2014). This feature likely influenced SM and IM, leading to a stronger preference for overt subjects and a higher

frequency of SBV structures compared to CMM. Furthermore, in the LTP annotation system, topic-comment structures are labelled as FOB, with their frequency being highest in CMM (0.77%), significantly higher than in SM (0.60%) and IM (0.64%). The Kruskal-Wallis test revealed a statistically significant difference in FOB frequency among the three variants ($H(2) = 51.485$, $p < 0.001$, $\eta^2 = .022$). In Chinese, topic sentences often involve the movement of syntactic constituents, and such movement-based structures typically give rise to long-distance syntactic dependencies (Phillips et al., 2005). Given the more frequent use of topic structures in CMM, this could be one of the reasons for its higher mean dependency distances. In contrast, SM and IM exhibit less frequent use of topicalization structures and a stronger preference for retaining overt subjects, thereby reducing long-distance dependencies. This may partially explain why their overall dependency distances are shorter than those of CMM. However, syntactic complexity is influenced by multiple factors, and the differences in topic structures are only one of the potential contributing factors. The specific reasons warrant further investigation in future studies.

6.2.4. *Simplification and Informal Trends as Explanations*

Trudgill (2011) posits that language varieties shaped by intense language contact typically display simplification tendencies. The analysis of coordinate (COO) dependency relations across the three Mandarin Chinese varieties reveals significant differences in their frequencies ($H(2) = 159.04$, $p < 0.001$, $\eta^2 = .066$). Specifically, SM and IM exhibit frequencies of 8.85% and 9.38%, both significantly lower than that of CMM (10.51%) ($ps < 0.001$). However, the difference between SM and IM does not reach statistical significance ($p = 0.093$). The results indicate significant differences between the two overseas Mandarin Chinese varieties and CMM. According to the LTP annotation framework, COO is primarily used to mark coordinate relations within complex sentences as well as coordinate structures at the phrasal level. The relatively lower frequency of COO in SM and IM suggests a variation from CMM in the syntactic organization of complex sentences and coordinate constructions. Specifically, SM and IM tend to break down coordinate complex sentences into multiple independent short clauses. This phenomenon may result from the influence of language contact environments, which encourage simplified syntactic structures that enhance communicative efficiency in multilingual contexts. To further validate this phenomenon, the frequency of HED dependency relations across the three Mandarin Chinese varieties is analysed. The primary function of HED is to mark the core predicate of a sentence, and its total frequency serves as an indirect indicator of the total number of sentences in a corpus. Statistical analysis reveals significant differences in HED frequencies among the three variants ($H(2) = 638.733$, $p < 0.001$, $\eta^2 = .199$). The significantly higher HED frequencies in

SM and IM compared to CMM ($ps < 0.001$) suggest that SM and IM contain more sentences than CMM in similarly sized corpora. When combined with the sentence length and distribution data, it is evident that the most frequent sentence lengths in SM and IM fall within shorter ranges, while CMM exhibits longer overall sentence lengths. These results further support the observation that SM and IM tend to organize information by breaking it into independent sentences rather than relying on complex sentences for integration. This structural preference reflects a tendency towards simplification and colloquiality, distinguishing these regional variants from CMM.

7. Conclusions

This study conducts a comparative analysis of the syntactic complexity across CMM, SM, and IM. The analysis reveals both differences and commonalities in the syntactic complexity of Mandarin Chinese in the Chinese mainland and in overseas Chinese communities.

First, the analysis of sentence length distribution demonstrates that all three Mandarin Chinese varieties conform to the EPNB model, indicating a universal preference for shorter sentences due to their lower cognitive cost and higher communicative efficiency. The fact that parameters p and α of the EPNB model effectively distinguish the three Mandarin Chinese varieties further highlights the differences among them. CMM exhibits the highest ASL, followed by SM, while IM shows the lowest. This trend is likely attributable to the influence of language policies, Chinese education, continuity of language heritage, and varying sociolinguistic environments. In particular, the standardized promotion of SM has resulted in a sentence length distribution intermediate between CMM and IM. Meanwhile, the historical restrictions in language policies led to a stronger tendency towards colloquialization and simplification in IM writing.

Second, CMM exhibits the highest MDD, followed by SM with an intermediate value, and IM with the lowest. The conservatism, simplification, and convergence observed in overseas Mandarin Chinese varieties can be attributed to prolonged exposure to local official languages, contact with southern Chinese dialects, and variations in language policies.

Finally, despite the distinct environments of these Mandarin Chinese varieties, they exhibit a significant degree of commonality, manifested in a universal trend towards dependency distance minimization. This trend is consistent with the principle of least effort in human language. However, this shared characteristic does not entirely eliminate the differences arising from geographical, sociocultural, and language contact factors, as evidenced by variations in the fitting parameters. This study finds that the parameters of the right-truncated modified Zipf-Alekseev distribution can also distinguish

between different Mandarin Chinese varieties, demonstrating possible typological universality.

Nevertheless, this study has certain limitations. It primarily focuses on editorial texts within the journalistic genre, which may constrain the universality of the findings, especially with regard to colloquial language use. Future research should incorporate a broader range of genres and registers (e.g. news reports, essays, novels) to better capture variation in Mandarin Chinese across contexts. Additionally, this study only investigates Mandarin Chinese varieties from the Chinese mainland, Singapore, and Indonesia, while global Mandarin exhibits a much wider spectrum of national and regional variations. To further validate the applicability of these findings, future studies should incorporate additional national and regional Mandarin Chinese varieties, including those spoken in the new Chinese immigrant communities. Moreover, expanding the analysis to consider a wider range of sociolinguistic factors may provide deeper insights into the linguistic and contextual influences on the syntactic complexity of Mandarin Chinese.

Notes

1. In this study, the term variety is used in an operational sense when CMM, SM, and IM are collectively referred to as ‘three varieties of Mandarin Chinese’. We do not suggest that CMM is a variety in the same sense as SM and IM. This formulation is adopted solely for linguistic and methodological purposes, as well as for convenience, and carries no political implications.
2. <http://livac.org/intro.php?lang=sc> (accessed 17 February 2025)
3. <http://huayu.languagetechnology.cn/> (accessed 17 February 2025)
4. <https://catalog.ldc.upenn.edu/LDC2009T14> (accessed 17 February 2025)
5. The Mandarin Chinese varieties editorial corpus constructed in this study was collected and compiled solely for academic research purposes. Due to third-party copyright concerns, the corpus is not publicly released in order to prevent large-scale dissemination of the original texts. Nevertheless, reasonable requests for access to copies for research purposes may be addressed to the authors.
6. <https://www.qiandaoribao.com/about-us/> (accessed 15 July 2025)
7. <https://www.qiandaoribao.com/about-us/> (accessed 15 July 2025)
8. <https://www.gcmcu.com/static/content/LMHY/HYGL/2023-0628/1123673240696291328.html> (accessed 15 July 2025)
9. <https://www.gcmcu.com/static/content/LMHY/HYGL/2023-06-28/1123670019013640192.html> (accessed 15 July 2025)
10. https://www.mfa.gov.cn/gjhdq_676201/gj_676203/yz_676205/1206_677076/1206x0_677078/ (accessed 15 July 2025)
11. https://www.mfa.gov.cn/gjhdq_676201/gj_676203/yz_676205/1206_677244/1206x0_677246/ (accessed 15 July 2025)
12. <https://github.com/HIT-SCIR/ltp> (accessed 1 February 2025) (accessed 15 July 2025)

13. Upon checking the corpus used in this study, we found that ellipsis marks did not occur in the data. Therefore, sentence segmentation based on periods, question marks, and exclamation marks was sufficient for our analysis.
14. The non-parametric Kruskal–Wallis test was employed because the data did not meet the assumption of normality, and this violation persisted even after logarithmic transformation. This rationale applies consistently across all relevant analyses in the manuscript.

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Disclosure Statement

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Appendix A.

Table A1. Definitions of Chinese major dependency relation types.

Tag	Description	Chinese Phrase	English Translation	Dependency Relation
ATT	attribute	piàoliang nǚhái (漂亮女孩)	beautiful girl	piàoliang <- nǚhái 漂亮 <- 女孩
ADV	adverbial	fēicháng měilì (非常美丽)	very beautiful	fēicháng <- měilì 非常 <- 美丽
VOB	verb-object	mǎi yì běn shū (买一本书)	buy a book	mǎi -> shū 买 -> 书
COO	coordinate	dàshān hé dàhǎi (大山和大海)	mountains and seas	dàshān -> dàhǎi 大山 -> 大海
RAD	right adjunct	piàoliang de nǚhái (漂亮的女孩)	a beautiful girl	piàoliang -> de 漂亮 -> 的
SBV	subject-verb	wǒ mǎi huā (我买花)	I buy flowers	wǒ <- mǎi 我 <- 买
POB	preposition-object	zài qiáng shàng (在墙上)	on the wall	zài -> shàng 在 -> 上
HED	head predicate	wǒ mǎi huā (我买花)	I buy flowers	mǎi <- ROOT 买 <- ROOT
LAD	left adjunct	píngguǒ hé xiāngjiāo (苹果和香蕉)	apples and bananas	hé <- xiāngjiāo 和 <- 香蕉
CMP	complement	zuò wán le gōngzuò (做完了工作)	finished doing work	zuò -> wán 做 -> 完
FOB	fronting-object	tā shénme shū dōu kàn (他什么书都看)	He reads every kind of book	shū <- kàn 书 <- 看
DBL	double-subject construction (pivot construction)	tā qǐng wǒ kàn diànyǐng (他请我看电影)	He invited me to watch a movie	qǐng -> wǒ 请 -> 我